

Summary

AI Robotics Engineer with an M.S. in Electrical Engineering, focused on ROS 2, Isaac Sim / Isaac Lab, and Isaac GR00T for robot control, VLA, and Digital Twin data workflows. Experienced in end-to-end integration across perception, data processing, policy inference, and simulation / real-world validation.

Key Results

3-4x automated data-collection throughput	70% → 90% cable-insertion success in 10 trials
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Experience

- AMTRAN TECHNOLOGY CO., LTD. | AI Robotics EngineerMay 2025 - Present | New Taipei, Taiwan
- Built ROS 2 control and simulation workflows for industrial and collaborative robot platforms with NVIDIA Isaac Sim / Isaac Lab, MoveIt 2, and cuMotion for motion planning, collision handling, pick-and-place, and cable tasks.
 - Delivered Isaac GR00T N1.7 data conversion, model training, and policy inference, separately validating virtual-data training followed by real-world inference and real-data training/inference workflows.
 - Integrated RealSense, YOLO OBB, AprilTag, and OpenCV with Quest 2 teleoperation and imitation-learning recording for virtual and physical robot data collection; automated data collection reached 3-4x manual operation in the same time window.
 - Owned robotics software from requirements and architecture through Docker deployment, testing, on-site integration, and maintenance; cable-insertion success improved from 70% to 90% in a 10-trial test.
- ISCOM | Software Engineer InternFeb 2022 - Jun 2022 | Kaohsiung, Taiwan
- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Robot Control & Digital Twin	ROS 2, NVIDIA Isaac Sim / Lab, MoveIt 2, cuMotion, virtual / physical robot workflows
VLA & Robot Learning	Isaac GR00T N1.7, VLA, Policy Inference, Imitation Learning, Robot State / Action
Perception & Visual Servoing	RealSense, YOLO OBB, AprilTag, OpenCV, Hand-Eye Calibration, Visual Servoing
Software, Embedded & IoT	Python, C++, FastAPI, React / TypeScript, Next.js, WebSocket, Docker, Linux, Arduino, ESP32, MQTT

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place